

Continuous planning: This involves continuous planning on what to implement based on technology evolution and dynamic business needs. It includes implementation of Agile principles using Scrum or Kanban methodologies

Continuous code inspection: This is static code analysis of the source code, where bugs and vulnerabilities are identified by tools. We can create our own rules as well, based on the flexibility provided by the tools. Over a period of time developers know what is correct and incorrect based on the rules configured. Such habits help over time to improve the quality of code.

Continuous integration: Continuous integration involves committing code to source code repositories as and when features are implemented or bugs are fixed. Merged code is verified against unit tests, based on predefined coverage, for better quality software. Code is compiled and a package is created.

Continuous delivery: Continuous delivery involves automated deployment of packages in the pre-production environment. The objective is to make package deployment ready.

Continuous testing: Continuous testing involves different types of automated testing such as functional testing, load testing, performance testing, security testing, and so on.

Continuous deployment: Continuous deployment uses the same automated deployment approach. However, deployment takes place in a production environment.

Cloud computing: Cloud resources are available on demand. The cloud helps to create a runtime environment in a short span of time.

Containers: Containers are very effective in resource utilisation and lightweight compared to virtual machines.

Infrastructure as a Code: Infrastructure as a Code involves scripting to create cloud or virtual resources, and manage those resources. There is no manual involvement. It is also known as software defined infrastructure. Multiple

configuration tools such as Chef, Puppet and Ansible are also used to create infrastructure or runtime environment. Various cloud service providers have also come up with offerings that allow the creation of multi-tier architecture in a specific cloud environment.

Configuration management: Configuration management tools such as Chef, Puppet and Ansible are used to change resources that are available in the cloud or non-cloud environments, to comply with the policy defined automatically without any manual intervention.

Continuous monitoring: Monitoring involves application and infrastructure monitoring. It is important to monitor resources and take corrective measures manually or in an automated way to fix issues.

Continuous notifications: Notifications on failures, success of operations, and capacity overload or under-utilisation are extremely important. These can be in the form of emails, SMSs, or Slack notifications based on the policies of the organisation or the business unit.

Continuous improvements: Automation and DevOps practices help to cultivate a mind set to continuously improve existing practices. Continuous improvements help to increase the quality of the product and the processes.

Continuous innovation: Continuous innovation takes place with the productivity gains realised by the developers. The time saved by carrying out these continuous practices can be invested in innovations, which are then integrated with existing processes.

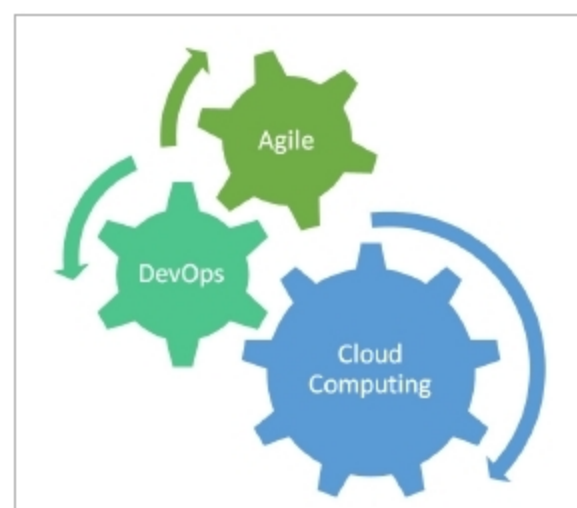


Figure 4: Agile DevOps and cloud computing

The implementation of continuous practices helps to bring a change in the existing organisation culture. It helps to change existing mind sets towards bottlenecks, and makes an organisation more efficient by saving time.

Now, after getting some understanding of DevOps and the cloud, let's understand how cloud and DevOps work together.

Cloud and DevOps

Agile principles serve as a driver for DevOps implementation, as there are shorter sprints for releases and automation helps to achieve the deadlines. Cloud computing is a great facilitator for implementing continuous practices, as resources are available in quick time for all the activities involved in application life cycle management.

Agile principles, cloud computing and DevOps practices enable faster growth in organisations. Let's understand how.

- With these practices, organisations are focused on agility, speed and quality at the same time.
 - Organisations understand the importance of disruptive innovation in technology. Even customers understand the need for innovation and efficiency.
- Cloud resources help to achieve faster time to market and agility, which helps to implement DevOps practices in a way that efficient outcomes are measured and realised.
- DevOps is all about cultural transformation.
 - With the economies of scale available with cloud resources, it is easier to provide the same type of resources for all environments; hence, all environment related issues due to inconsistencies in the resources are avoided. Project teams face all issues in a development environment rather than facing them later; hence, issues can be fixed earlier.
- Knowledge transfers happen between the development and operations teams. These teams understand each other's problems and work together.